

DABI — Practical 11

**Data Cleaning, Preprocessing and
Dashboard Design Planning**

Coal Resource Intelligence System (CRIS)

Group No	Group 11
Roll No	161
Name	Tanu Sharma
Subject	Data Analytics and Business Intelligence (DABI)
Practical No	11

1. Dataset Selected

Dataset Information

Dataset Name	Coal Resource Intelligence System (CRIS)
Number of Datasets	3 Official Datasets
Source	NDAP — National Data and Analytics Platform
Ministry	Ministry of Coal, Government of India
Source Link	https://ndap.gov.in
Time Period	Financial Year 2006 — 2020
Total Records	9,235 rows across all 3 datasets

Datasets Overview

Dataset	Rows	Columns	Key Fields
Coal Consumption	2,988	6	State, Year, Sector, Coal_Consumed_MT
Coal Supply Production	830	8	State, Company, Supply_Sector, Coal_Supplied_MT
Coal Reserves Production	5,417	8	Coal_Field, Reserve_Type, Depth, Coal_Reserved_MT

2. Dataset Issues Found

Issues Identified During Data Inspection

#	Issue Type	Description
1	Long Column Names	Columns like 'Coal Consumed (UOM:t(Tonne)), Scaling Factor:1000000' were extremely long and unusable in DAX formulas
2	Year Format — Text	Year column stored as 'Financial Year (Apr - Mar), 2020' instead of numeric 2020 — could not be used in time-series charts
3	Inconsistent Casing	Reserve_Type column had both 'PROVED' and 'Proved', 'INDICATED' and 'Indicated' — causing duplicates in filtering
4	Missing Depth Midpoint	Depth column had ranges like '0-300', '300-600' but no numeric midpoint for weighted calculations
5	Redundant Country Column	Country column contained only 'India' for all 9,235 rows — added no analytical value
6	Typo in Column Name	Column 'Coal Supplyling Sector' had a typo ('Supplyling' instead of 'Supplying')
7	No Relationships	Three datasets had no pre-defined relationships — required manual linking via State and Year fields

3. Steps Taken to Clean Data

All cleaning was performed in Power Query Editor (Home → Transform Data) in Power BI Desktop.

Step 1: Rename Long Column Names (All 3 Tables)

Double-clicked each column header and renamed to short clean names:

- Coal Consumed (UOM:t(Tonne)), Scaling Factor:1000000 → Coal_Consumed_MT
- Coal Supplied (UOM:t(Tonne)), Scaling Factor:1000000 → Coal_Supplied_MT
- Coal Reserved Quantity (UOM:t(Tonne)), Scaling Factor:1000000 → Coal_Reserved_MT
- Coal Supplying Company Name → Company
- Coal Consuming Sectors → Sector
- Coal Supplying Sector → Supply_Sector (also fixed typo)
- Name Of The Coal Field → Coal_Field
- Depth Of The Reserve → Depth
- Type Of Reserve → Reserve_Type
- Types Of Coal → Coal_Type_Con

Step 2: Fix Year Column — Text to Number (All 3 Tables)

Year column contained 'Financial Year (Apr - Mar), 2020'. Steps taken:

- Selected Year column in each table
- Transform tab → Extract → Text After Delimiter → typed ', ' (comma space)
- Result: extracted just '2020', '2019', '2006' etc.
- Changed data type to Whole Number (ABC → 123)
- Repeated for all 3 tables

Step 3: Fix Reserve_Type Casing (Reserves Table Only)

Reserve_Type had inconsistent uppercase values. Steps taken:

- Selected Reserve_Type column
- Transform → Format → Capitalize Each Word
- Replace Values: 'PROVED' → 'Proved'
- Replace Values: 'INDICATED' → 'Indicated'
- Replace Values: 'INFERRED' → 'Inferred'
- All values now consistent for accurate filtering

Step 4: Add Depth_Midpoint Column (Reserves Table)

Added a new numeric column for weighted depth calculations. Add Column → Conditional Column:

- If Depth = '0-300' Then 150
- If Depth = '300-600' Then 450
- If Depth = '600-1200' Then 900

- If Depth = '0-600' Then 300
- Else 300
- Changed type to Whole Number

Step 5: Remove Country Column (All 3 Tables)

Country column had only 'India' in all rows — no analytical value.

- Right-clicked Country column header in each table
- Selected 'Remove' to delete column
- Repeated for all 3 tables
- Saved 1 column per table = cleaner model

4. New Columns Created

Column Name	Table	Formula / Logic
Depth_Midpoint	Coal_Reserves	Conditional Column in Power Query: If Depth = '0-300' → 150 (midpoint) If Depth = '300-600' → 450 (midpoint) If Depth = '600-1200' → 900 (midpoint) If Depth = '0-600' → 300 (midpoint) Else → 300 <i>Purpose: Enables weighted average depth calculation for mining cost analysis</i>

DAX Measures Created

Measure Name	DAX Formula
Total_Consumption_MT	SUM(Coal_Consumption[Coal_Consumed_MT])
Total_Supplied_MT	SUM(Coal_Supply[Coal_Supplied_MT])
Supply_Ratio	DIVIDE([Total_Supplied_MT], [Total_Consumption_MT], 0)
Depletion_Index	DIVIDE(CALCULATE(SUM(Coal_Reserves[Coal_Reserved_MT]), Coal_Reserves[Reserve_Type]="Proved"), SUM(Coal_Reserves[Coal_Reserved_MT]), 0) * 100
Company_Yield_Pct	DIVIDE(SUM(Coal_Supply[Coal_Supplied_MT]), CALCULATE(SUM(Coal_Supply[Coal_Supplied_MT]), ALLEXCEPT(Coal_Supply, Coal_Supply[State])), 0) * 100
Avg_Depth_Factor	DIVIDE(SUMX(Coal_Reserves, Coal_Reserves[Depth_Midpoint] * Coal_Reserves[Coal_Reserved_MT]), SUM(Coal_Reserves[Coal_Reserved_MT]), 0)
Total_Reserves_MT	SUM(Coal_Reserves[Coal_Reserved_MT])

5. Dashboard Layout

The dashboard is designed across 4 pages with a consistent purple theme. Each page focuses on a specific analytical area of coal resource intelligence.

Page 1 — Executive Summary

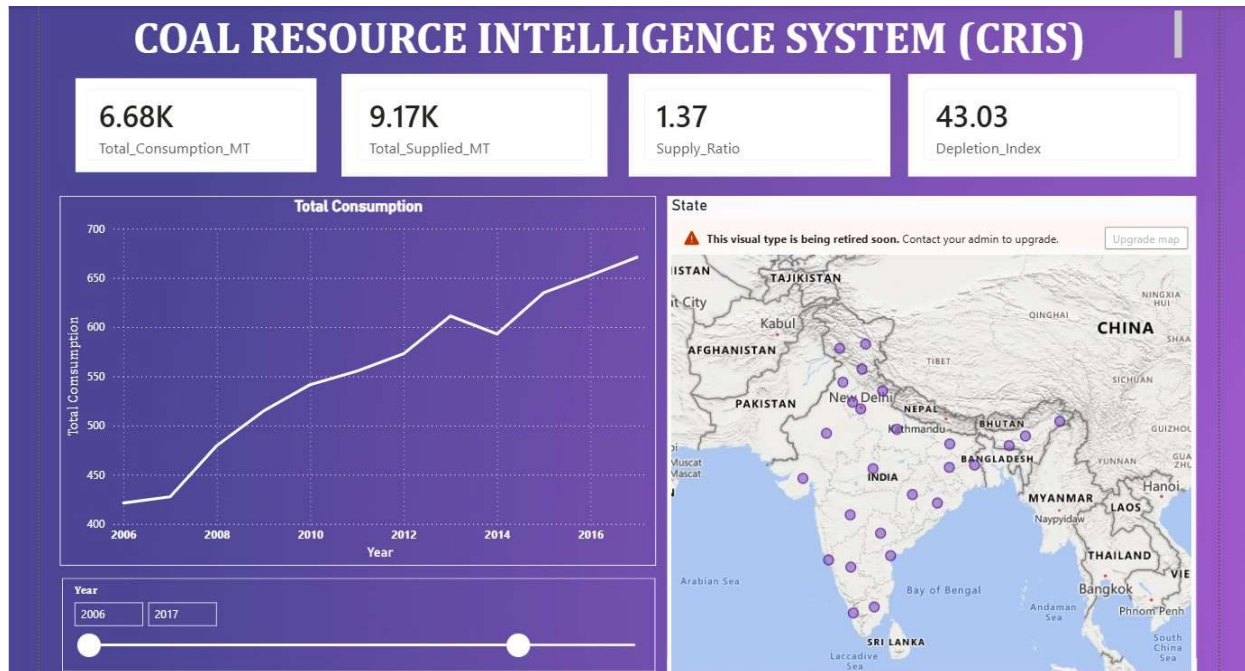


Figure 1: Executive Summary — KPI Cards, Consumption Trend, India Map

Page 2 — Consumption Analysis

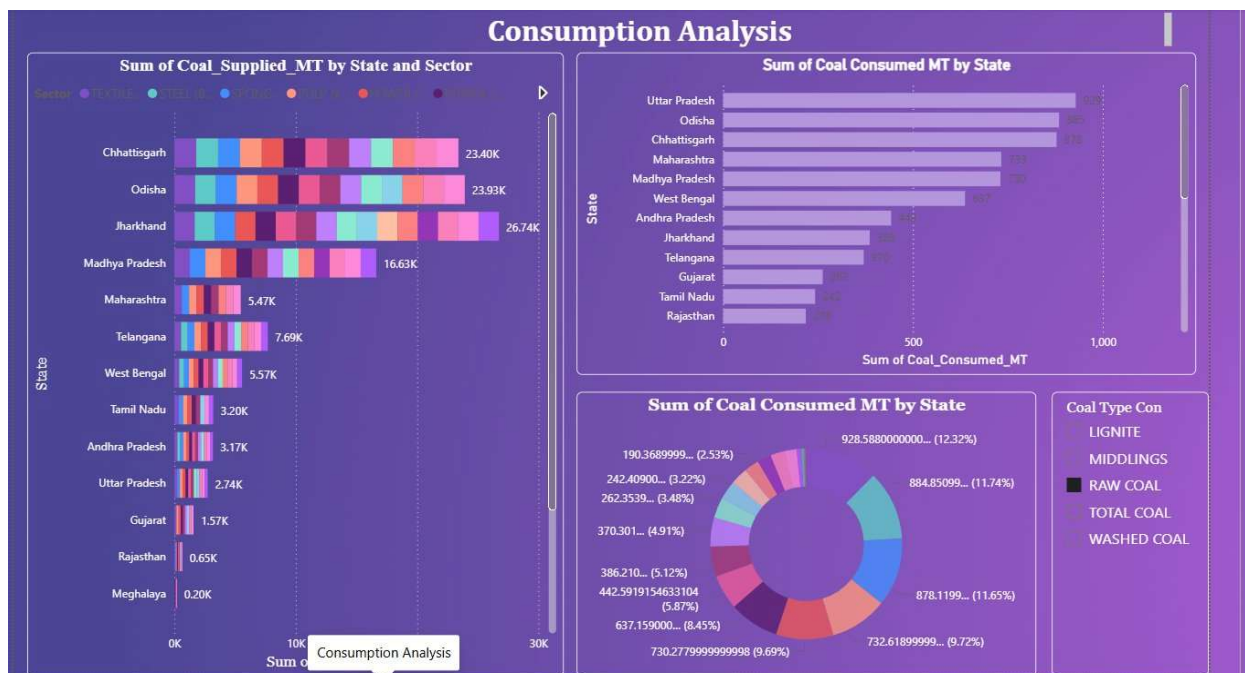


Figure 2: Consumption Analysis — State-wise Sector Breakdown

Page 3 — Supply Analysis

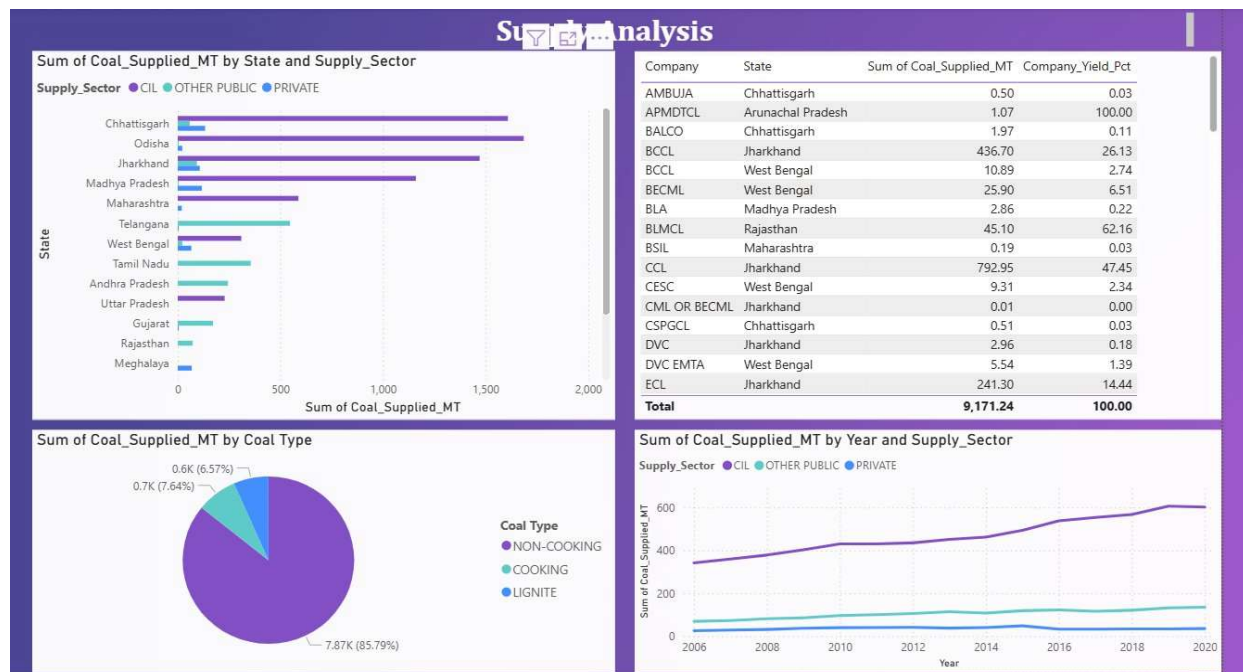


Figure 3: Supply Analysis — Company Table, Coal Type Distribution

Page 4 — Reserves & Sustainability



Figure 4: Reserves & Sustainability — Coal Fields, Depletion Gauge, Treemap

6. Charts Used

Page	Chart Type	Data Fields Used
Page 1	KPI Card (×4)	Total_Consumption_MT, Total_Supplied_MT, Supply_Ratio, Depletion_Index
Page 1	Line Chart	Year (X-axis) vs Total_Consumption_MT (Y-axis)
Page 1	Map Visual	State (Location) vs Total_Consumption_MT (Bubble size)
Page 1	Slicer	Year range filter (2006–2020)
Page 2	Stacked Bar Chart	State (Y-axis), Coal_Supplied_MT (X-axis), Sector (Legend)
Page 2	Clustered Bar Chart	State (Y-axis), Coal_Consumed_MT (X-axis)
Page 2	Donut Chart	Sector (Legend), Coal_Consumed_MT (Values)
Page 2	Slicer	Coal_Type_Con filter
Page 3	Clustered Bar Chart	State (Y-axis), Coal_Supplied_MT (X-axis), Supply_Sector (Legend)
Page 3	Table	Company, State, Coal_Supplied_MT, Company_Yield_Pct
Page 3	Pie Chart	Coal Type (Legend), Coal_Supplied_MT (Values)
Page 3	Line Chart	Year (X-axis), Coal_Supplied_MT (Y-axis), Supply_Sector (Legend)
Page 4	Stacked Bar Chart	Coal_Field (Y-axis), Coal_Reserved_MT (X-axis), Reserve_Type (Legend)
Page 4	Gauge	Depletion_Index (Value), 0 (Min), 100 (Max)
Page 4	Clustered Bar Chart	Depth (Y-axis), Coal_Reserved_MT (X-axis)
Page 4	Treemap	State (Category), Coal_Field (Details), Coal_Reserved_MT (Values)

7. Why These Charts

Chart Type	Reason for Selection
KPI Cards	Display single most-important numbers at a glance. Used for Total Consumption, Supply, Ratio and Depletion Index so decision-makers see key metrics immediately without searching through charts.
Line Chart	Best for showing trends over time. Used for Consumption and Supply trends from 2006–2020 to clearly visualize if coal usage is increasing or decreasing year by year.
Map Visual	Geographical distribution is best understood on a map. Bubble size shows which states consume or produce most coal. Gives instant spatial understanding of regional patterns.
Stacked Bar Chart	Shows both total value AND composition simultaneously. Used for State vs Sector breakdown so viewer sees total consumption per state AND which sectors (Power, Cement, etc.) contribute most.
Clustered Bar Chart	Best for comparing values across multiple categories side by side. Used for state-wise supply comparison with CIL, PRIVATE and OTHER PUBLIC sectors visible simultaneously.
Donut Chart	Shows proportional distribution elegantly. Used for Sector share and Coal Type split (NON-COOKING 85.79%, COOKING 7.64%, LIGNITE 6.57%) — easy to see dominant category.
Pie Chart	Ideal for showing part-to-whole relationship when categories are limited (3 coal types). Viewer instantly grasps that NON-COOKING coal dominates Indian supply.
Table Visual	Used for Company-wise data because exact numbers matter for 60 companies. Charts would make individual values unreadable at this volume.
Gauge Visual	Speedometer-style gauge is intuitive for index values with a known range. Depletion Index of 42.98% on a 0–100 scale is immediately understood — closer to 100 = higher risk.
Treemap	Hierarchical data (State → Coal Field → Quantity) is best shown as nested rectangles. Bigger box = more reserves. Viewer understands Jharkhand and Odisha dominate reserves instantly.
Slicer	Interactive filter lets users drill into specific years or coal types without rebuilding charts. Makes dashboard dynamic and self-service for any stakeholder.

Conclusion

This practical demonstrates a complete ETL and BI pipeline — from raw government data to a professional interactive dashboard. The CRIS dashboard integrates 9,235 rows across 3 official NDAP datasets, applies rigorous data cleaning in Power Query, creates 7 DAX measures, and delivers 4 dashboard pages with 16 visualizations covering coal consumption, supply, and reserve sustainability across 26 Indian states.